NAVRACHANA UNIVERSITY

Teaching Plan

Natural Language Processing (CSE704)

Credit : 2

Faculty	<table><tr><th>Name</th><th>Contact</th><th>Office Hours</th><th>Office Location</th></tr><tr><td>Jaideepsinh Raulji</td><td>jaideep.raulji@nuv.ac.in</td><td></td><td></td></tr></table>
School	SET
Semester	Autumn (July to Dec 2026)
Lecture time & Weedays	<table><tr><td>SECTION_1</td><td>Thursday</td><td>11:00:00-12:00:00</td><td>06-07-2026 TO 12-07-2026</td></tr><tr><td></td><td>Friday</td><td>10:00:00-11:00:00 12:00:00-13:00:00</td><td>06-07-2026 TO 12-07-2026 13-07-2026 TO 06-11-2026</td></tr><tr><td></td><td>Wednesday</td><td>16:00:00-17:00:00 12:00:00-13:00:00 12:00:00-13:00:00</td><td>13-07-2026 TO 02-08-2026 03-08-2026 TO 09-08-2026 10-08-2026 TO 06-11-2026</td></tr></table>
Pre-requisite	
Co-requisite	
Anti-requisite	
Course Description	Natural Language Processing (NLP) or Computational Linguistics is gaining popularity due to better computational facilities available on personal computers and mobile systems. None of the single natural language dominates the world. To cater to the need of the scenario, study of NLP cannot be overlooked. Applications of NLP are pervasive, because people communicate through spoken and written language hence the need for NLP and its peripheral knowledge is evident. The course is designed to expand knowledge about NLP, Language Analysis and Modeling Techniques, studying nature of Indian languages, Morphology, Semantics, Parsing, Disambiguation, Hybrid techniques of processing, Basics of Machine Translation, Information Retrieval, Text Summarization etc.
Course Abstract	-
Course Objectives	Introduce students to 1. Language Dynamics 2. To understand deviations between Natural and Formal languages. 3. Grammar and Language understanding and modeling. 4. Morphology, Parsing Techniques, ML and DL for Language processing. 5. Analysis and Implementation of NLP based algorithms and applications.
Course Outcomes	CO1. Basics understanding of NLP elements, scripts, speech, divergence pattern and morphological functions. CO2. NLP element processing, encodings, regular expression, tokenization, and pattern matching. CO3. Implementing different vectorization methods, Machine Learning and Deep Learning model for NLP. CO4. Implementation of sequence learning through RNN, LSTM, Sequence to Sequence, transformer-based model. CO5. Basic understanding of small and large language models.
Course Typology	Theory
Course Outline	**Introduction to Natural Language Processing (NLP)** Natural Language Understanding, Natural Language Generation, History of natural languages and written scripts, Natural Language Genesis, Divergence in different language categories. **Language Structure** Structure of word, Tokens, Lexemes, Morphemes, Ambiguity, Regular Expression, Automata, Finite State Machines, Transducer, String Regex Patterns, Tokenizers, Word Order and Grammar. **Elements of Natural Language and Preprocessing Text** Knowing characters, Unicode and UTF-8, Transliteration, Romanization, Tokens, Types of Tokenizers, Stopwords, Stemming and Lemmatization, Normalization of Text, Morphology, Parts-of-Speech Tagging, Syntactic Parsing, Semantic Parsing, Language Modelling, Probabilistic Modelling, Vectorization, BoW, TF-IDF, N-Grams, One-Hot Encoding, Label Encoding, Zips Law. **Machine Learning for Natural Language Processing** Feature Engineering ,Naïve Bayes Classifier, Tree Based Models/Classifier, Linear and Logistic Regression, Support Vector Machine, Deep Learning for NLP, Word Embeddings (Word2Vec, Glove, FastText) **Sequence Processing with Deep Learning** Recurrent Neural Network (RNN), Long Short Term Memory (LSTM), Sequence to Sequence Learning, Stackable Deep Learning for NLP, Transformers, BERT, Introduction to Large Language Models. Applications of NLP, Text Classification, Sentiment Analysis, Summarization, Translation etc.
Pedagogy	Experiential Learning
Expectations from Students	Pre-reads, pre-Lab work before joining the session, Referring research papers pertaining to NLP
Assessment / Evaluation	
Project / Assignment Details	Group Projects and Assignments will be shared with students.
Course Material	Main Reference Books: 1. Hobson Lane, Maria Dyhshel, Natural Language Processing in Action, Second Edition, Manning Publications. 2. Daniel Jurafsky, James Martin, “Speech and Language Processing” – Pearson Education. 3. Daniel Bikel, Imed Zitouni, “Multilingual Natural Language Processing Applications”, Pearson Education, First Edition 2013. 4. Ian Goodfellow, Yoshua Bengio, and Aaron Courville. 2016. Deep Learning. The MIT Press. 5. Additional Reference Books/Resources: 1. Taweh Beysolow, “Applied Natural Language Processing with Python”, Apress pubication. 2. Sebastian Raschka, Vahid Mirjalili, “Python Machine Learning”, Third Edition, Packt Publishing. 3. Akshay Kulkarni, Adarsha Sivananda, et al, “Applied Generative AI for Beginners”, Apress publications. 4. Akshar Bharati, Vineet Chaitanya, Rajeev Sangal, “Natural Language Processing : A Paninian Perspective”, PHI Learning. 5. Kamal Rawat, Meenakshi, “Dynamic Programming for Coding Interviews”, Notion Press, First Edition 2017. 6. https://github.com/rasbt/LLMs-from-scratch 7. https://jalammar.github.io/illustrated-transformer/ 8. https://poloclub.github.io/transformer-explainer/ 9. https://huggingface.co/transformers/v3.3.1/pretrained_models.html
Additional Information	This is an elective course.

Session Plan

Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
Introduction to Natural Language Processing (NLP)	1	Natural Language Understanding, Natural Language Generation, History of natural languages and written scripts	Jurafsky and Martin, "Speech and Language Processing", Ch 1, 16	Lecturing and Class Discussions	-
Introduction to Natural Language Processing (NLP)	2	Natural Language Understanding, Natural Language Generation, History of natural languages and written scripts	Jurafsky and Martin, "Speech and Language Processing", Ch 1, 16	"	-
Introduction to Natural Language Processing (NLP)	3	Natural Language Genesis, Divergence in different language categories.	Jurafsky and Martin, "Speech and Language Processing", Ch 1, 16	Lecturing, Discussions and Application Demo	-
Language Structure	4	Structure of word, Tokens, Lexemes, Morphemes, Ambiguity,	Jurafsky and Martin, "Speech and Language Processing", Ch 1, 2, 4, 16	"	-
Language Structure	5	Regular Expression, Automata, Finite State Machines, Transducer, String Regex Patterns,	Jurafsky and Martin, "Speech and Language Processing", Ch 1, 2, 4, 12	"	-
Language Structure	6	Tokenizers, Word Order and Grammar.	Jurafsky and Martin, "Speech and Language Processing", Ch 1, 2, 4, 12, Hobson Lane et. al, "Natural Language Processing in Action", Ch. 2,3	Lecturing, Discussions and Application Demo	-
Elements of Natural Language and Preprocessing Text	7	Knowing characters, Unicode and UTF-8, Transliteration, Romanization, Tokens, Types of Tokenizers	Jurafsky and Martin, "Speech and Language Processing", Ch 1, 2, 4	"	-
Elements of Natural Language and Preprocessing Text	8	Stopwords, Stemming and Lemmatization	Jurafsky and Martin, "Speech and Language Processing", Ch 1, 2, 4	"	-
Elements of Natural Language and Preprocessing Text	9	Normalization of Text, Morphology, Parts-of-Speech Tagging, Syntactic Parsing, Semantic Parsing, Language Modelling	Jurafsky and Martin, "Speech and Language Processing", Ch 5	"	-
Elements of Natural Language and Preprocessing Text	10	Normalization of Text, Morphology, Parts-of-Speech Tagging, Syntactic Parsing, Semantic Parsing, Language Modelling	Jurafsky and Martin, "Speech and Language Processing", Ch 5, 14	"	-
Elements of Natural Language and Preprocessing Text	11	Probabilistic Modelling, Vectorization	Jurafsky and Martin, "Speech and Language Processing", Ch 5, 6, 14	"	-
Elements of Natural Language and Preprocessing Text	12	BoW, TF-IDF, N-Grams, One-Hot Encoding, Label Encoding, Zips Law.	Jurafsky and Martin, "Speech and Language Processing", Ch 5, 6, 14, Hobson Lane et. al, "Natural Language Processing in Action", Ch. 3, 4	"	-
Elements of Natural Language and Preprocessing Text	13	BoW, TF-IDF, N-Grams, One-Hot Encoding, Label Encoding, Zips Law.	Jurafsky and Martin, "Speech and Language Processing", Ch 5, 6, 14	"	-
Machine Learning for Natural Language Processing	14	Feature Engineering ,Naïve Bayes Classifier, Tree Based Models/Classifier	Jurafsky and Martin, "Speech and Language Processing", Ch 15	"	-
Machine Learning for Natural Language Processing	15	Feature Engineering ,Naïve Bayes Classifier, Tree Based Models/Classifier	Jurafsky and Martin, "Speech and Language Processing", Ch 15	"	-
Machine Learning for Natural Language Processing	16	Feature Engineering ,Naïve Bayes Classifier, Tree Based Models/Classifier	Jurafsky and Martin, "Speech and Language Processing", Ch 15	"	-
Machine Learning for Natural Language Processing	17	Linear and Logistic Regression, Support Vector Machine,	Hobson Lane et. al, "Natural Language Processing in Action", Ch. 5	"	-
Machine Learning for Natural Language Processing	18	Linear and Logistic Regression, Support Vector Machine,	Hobson Lane et. al, "Natural Language Processing in Action", Ch. 5	"	-
Machine Learning for Natural Language Processing	19	Deep Learning for NLP	Hobson Lane et. al, "Natural Language Processing in Action", Ch. 5	"	-
Machine Learning for Natural Language Processing	20	Word Embeddings (Word2Vec, Glove, FastText)	Hobson Lane et. al, "Natural Language Processing in Action", Ch. 6	"	-
Machine Learning for Natural Language Processing	21	Word Embeddings (Word2Vec, Glove, FastText)	Hobson Lane et. al, "Natural Language Processing in Action", Ch. 6	"	-
Sequence Processing with Deep Learning	22	Recurrent Neural Network (RNN), Long Short Term Memory (LSTM), Sequence to Sequence Learning	Hobson Lane et. al, "Natural Language Processing in Action", Ch. 9, 10	"	-
Sequence Processing with Deep Learning	23	Recurrent Neural Network (RNN), Long Short Term Memory (LSTM), Sequence to Sequence Learning	Hobson Lane et. al, "Natural Language Processing in Action", Ch. 5	"	-
Sequence Processing with Deep Learning	24	Recurrent Neural Network (RNN), Long Short Term Memory (LSTM), Sequence to Sequence Learning	Hobson Lane et. al, "Natural Language Processing in Action", Ch. 9, 10	"	-
Sequence Processing with Deep Learning	25	Stackable Deep Learning for NLP	Hobson Lane et. al, "Natural Language Processing in Action", Ch. 9, 10	"	-
Sequence Processing with Deep Learning	26	Transformers, BERT	Refer https://jalammar.github.io/illustrated-transformer/	"	-
Sequence Processing with Deep Learning	27	Transformers, BERT	Refer https://jalammar.github.io/illustrated-transformer/	"	-
Sequence Processing with Deep Learning	28	Introduction to Large Language Models. Applications of NLP, Text Classification, Sentiment Analysis, Summarization, Translation etc.	Refer https://github.com/rasbt/LLMs-from-scratch	"	-
Sequence Processing with Deep Learning	29	Introduction to Large Language Models. Applications of NLP, Text Classification, Sentiment Analysis, Summarization, Translation etc.	Refer https://github.com/rasbt/LLMs-from-scratch	"	-
Revision	30	Revision	--	"	-

Publish Status	Published